

Deploying Web App on ECS – Step 1: Configure Network

To Begin with the Lab

Summary of the Lab

In this step, a secure networking foundation was built for the ECS project using **Amazon VPC**. A new VPC was created after removing the default VPC to keep the AWS environment clean and avoid configuration conflicts. Four subnets were then created across two Availability Zones, consisting of two public subnets and two private subnets, followed by the creation of an **Internet Gateway**, a **NAT Gateway**, and separate public and private route tables. Finally, the correct subnet associations and routing rules were configured so that public resources could access the internet directly while private ECS tasks could securely access AWS services through the NAT Gateway. This completed the network infrastructure required before configuring security groups and deploying the application.

Understanding Amazon VPC for Secure ECS Networking

Amazon VPC is an isolated virtual network that allows you to securely host AWS resources while controlling how they communicate with each other and with the internet. It exists so that public-facing components, such as an **Application Load Balancer**, can remain accessible from the internet, while sensitive resources like **Amazon ECS** tasks stay protected inside private subnets. A **NAT Gateway** allows resources in private subnets to access external AWS services without exposing them to inbound internet traffic. This design improves both security and high availability by distributing resources across multiple Availability Zones.

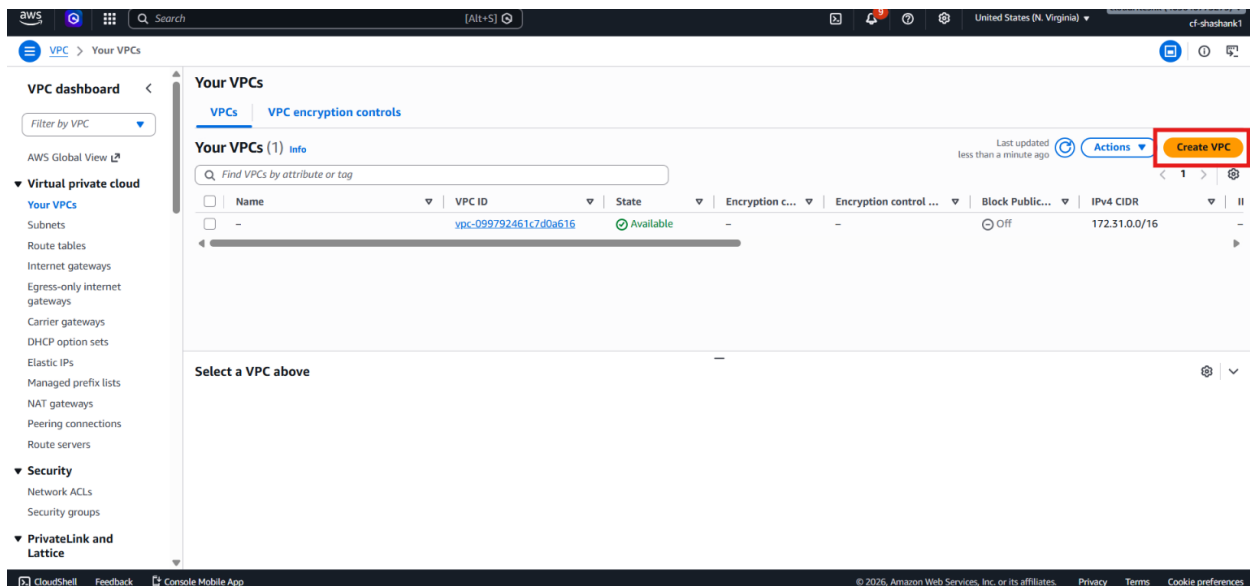
Example:

An ECS application runs two tasks inside private subnets in **ap-south-1a** and **ap-south-1b**, while an **Application Load Balancer** runs in two public subnets. Users access the application through the load balancer, and the ECS tasks use the **NAT Gateway** to pull Docker images from **Amazon ECR** and upload files to **Amazon S3**, without receiving public IP addresses.

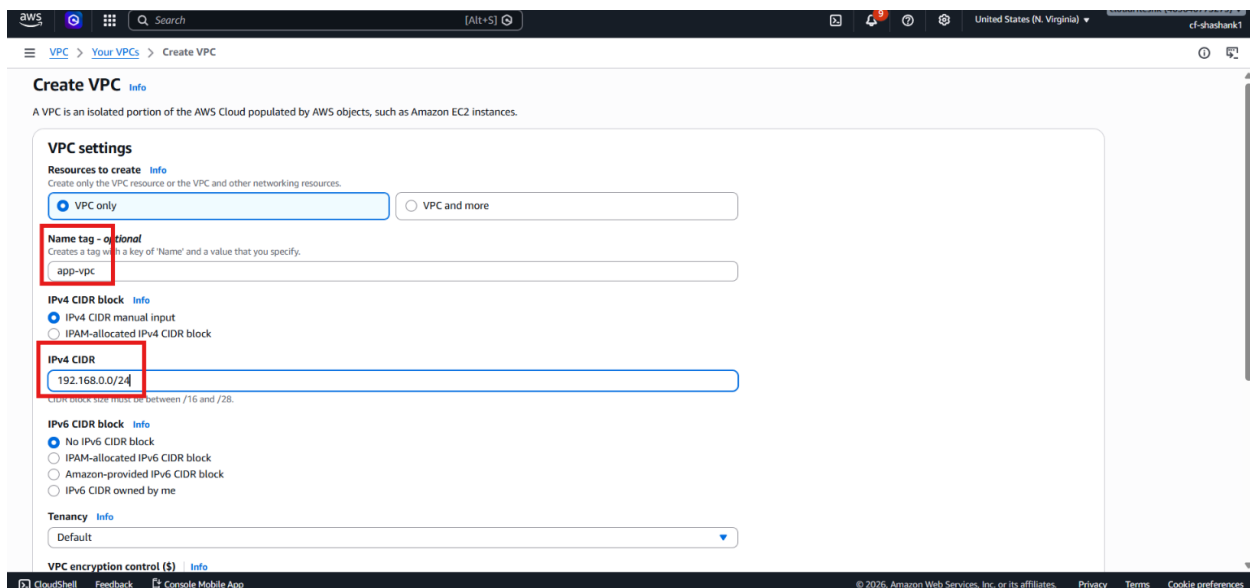
Lab Steps

Step 1 – Create the Amazon VPC

- Sign in to the **AWS Management Console**.
- Open **Amazon VPC**.
- Click **Create VPC**.

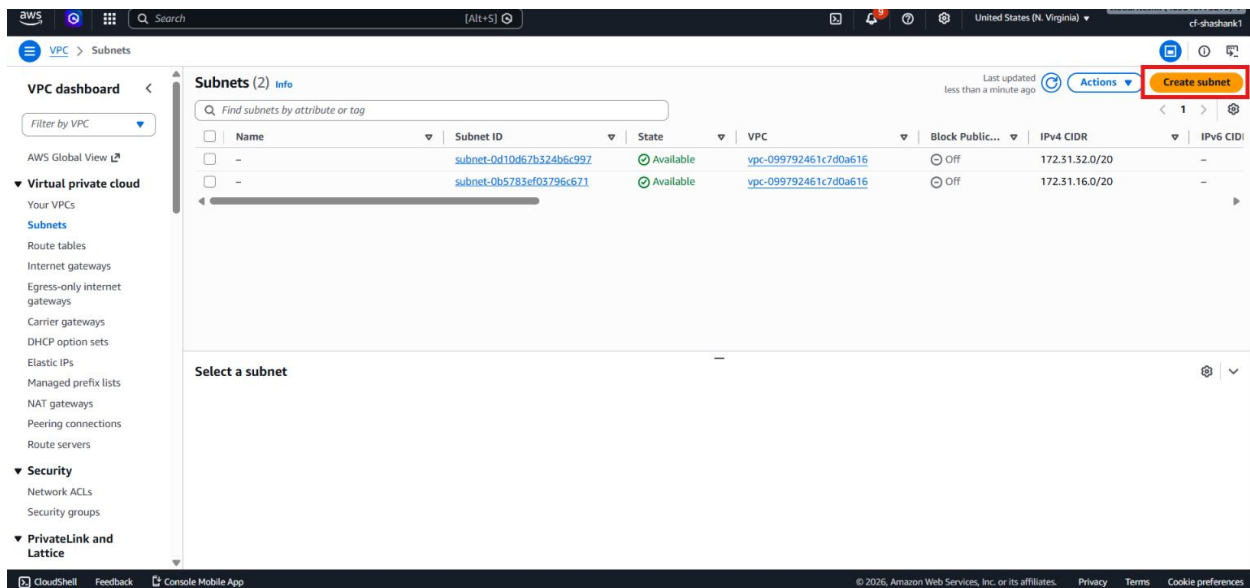


- Enter the VPC name.
- Set the IPv4 CIDR block to 192.168.0.0/24.
- Click **Create VPC**.

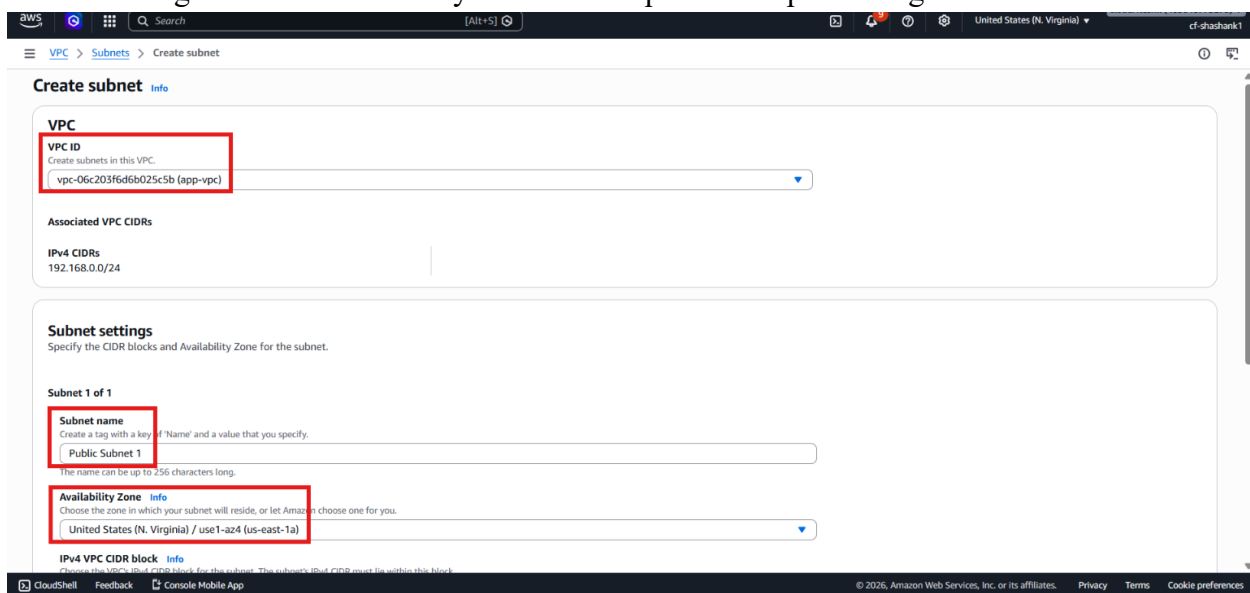


Step 2 – Create the Public and Private Subnets

- Open **Subnets**.
- Click **Create subnet**.



- Select the newly created VPC.
- I am using us-east-1 so that why I selected its public and private regions.



- Create **Public Subnet 1** in **ap-south-1a** with **IP Address Block**: 192.168.0.0/26.
- Create **Public Subnet 2** in **ap-south-1b** with **IP Address Block**: 192.168.0.64/26.

Subnet 2 of 4

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public Subnet 2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1b

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.64/26 64 IPs

Tags - optional

Key	Value - optional
Name	Public Subnet 2

[Add new tag](#)
You can add 49 more tags.
[Remove](#)

- Create **Private Subnet 1** in **ap-south-1a** with **IP Address Block: 192.168.0.128/26**.

Subnet 3 of 4

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private Subnet 1
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.128/26 64 IPs

Tags - optional

Key	Value - optional
Name	Private Subnet 1

[Add new tag](#)
You can add 49 more tags.
[Remove](#)

- Create **Private Subnet 2** in **ap-south-1b** with **IP Address Block: 192.168.0.192/26**.
- Verify that all four subnets are created successfully.

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private Subnet 2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1b

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.192/26 64 IPs

Tags - optional
Key Value - optional
Name Private Subnet 2 Remove
Add new tag
You can add 49 more tags.
Remove
Add new subnet

Cancel Create subnet

Step 3 – Create and Attach the Internet Gateway

- Open Internet Gateways.
- Click Create Internet Gateway.

Internet gateways (1) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
default-IG	igw-0e69daf64d15546cd	Attached	vpc-099792461c7d0a616	463646775279

Select an internet gateway above

Create internet gateway

- Enter the gateway name: MyIGW
- Click Create Internet Gateway.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

VPC > Internet gateways > Create internet gateway

Create internet gateway info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.
MyIGW

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
MyIGW

Remove

Add new tag
You can add 49 more tags.

Cancel Create internet gateway

https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1 © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

- Click **Attach to VPC**.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

VPC > Internet gateways > igw-0390e723e8860f7d3

VPC dashboard < Filter by VPC AWS Global View

Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only internet gateways Carrier gateways DHCP option sets Elastic IPs Managed prefix lists NAT gateways Peering connections Route servers

Security Network ACLs Security groups

PrivateLink and Lattice

The following internet gateway was created: igw-0390e723e8860f7d3 - MyIGW. You can now attach to a VPC to enable the VPC to communicate with the internet. Attach to a VPC

igw-0390e723e8860f7d3 / MyIGW

Actions

Details info

Internet gateway ID igw-0390e723e8860f7d3	State Detached	VPC ID -	Owner 463646775279
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Tags (1)

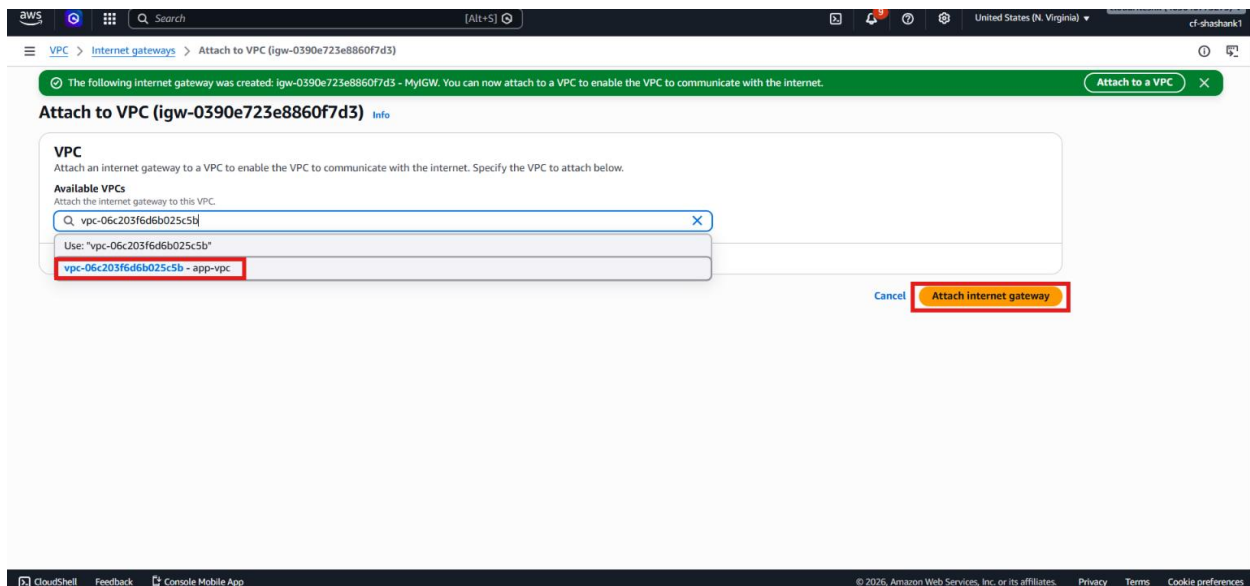
Search tags

Key	Value
Name	MyIGW

Manage tags < 1 >

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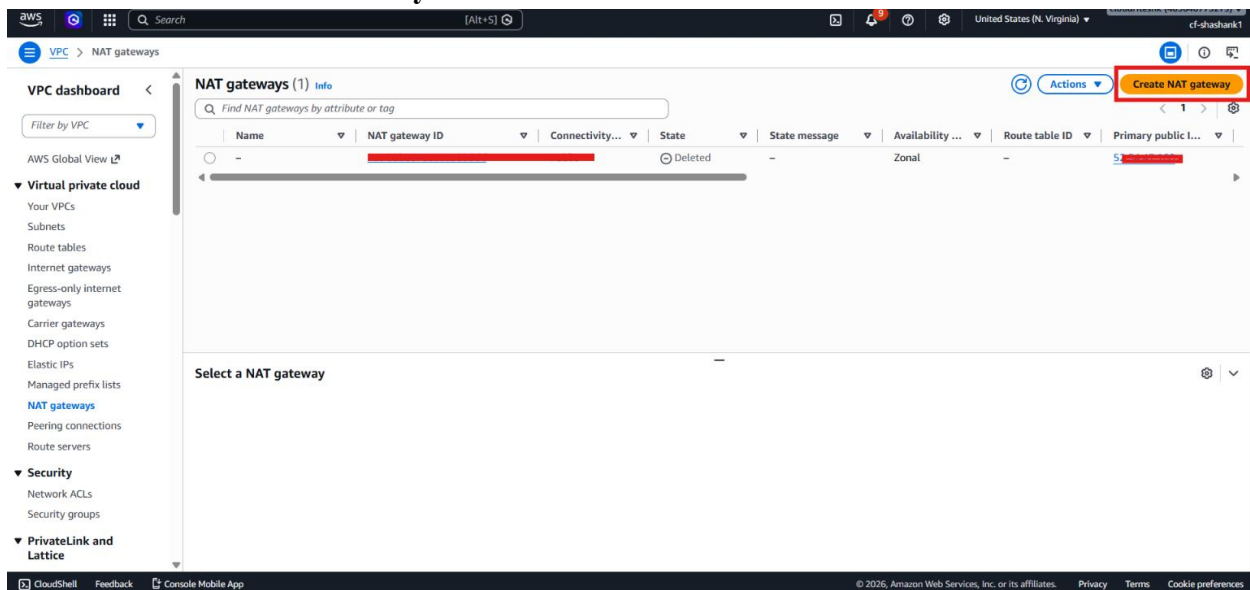
- Select the newly created VPC.



- Attach the Internet Gateway successfully.

Step 4 – Create the NAT Gateway

- Open NAT Gateways.
- Click Create NAT Gateway.



- Enter the NAT Gateway name: MyNGW
- Select one of the public subnets.
- Allocate a new Elastic IP.
- Click Create NAT Gateway.

AWS [Search] [Alt+S] United States (N. Virginia) cf-shashank1

VPC > NAT gateways > Create NAT gateway

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

MyNGW

The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
 ☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-018f92fe3e08eca78 (Public Subnet 1)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

Select an Elastic IP

[Allocate Elastic IP](#)

▶ Additional settings [Info](#)

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- Wait until the NAT Gateway status changes to **Available**.

Step 5 – Create the Public Route Table

- Open Route Tables.
- Click **Create Route Table**.

AWS [Search] [Alt+S] United States (N. Virginia) cf-shashank1

VPC > Route tables

VPC dashboard

Filter by VPC

AWS Global View
 Virtual private cloud
 Your VPCs
 Subnets
 Route tables
 Internet gateways
 Egress-only internet gateways
 Carrier gateways
 DHCP option sets
 Elastic IPs
 Managed prefix lists
 NAT gateways
 Peering connections
 Route servers
 Security
 Network ACLs
 Security groups
 PrivateLink and Lattice

Route tables (1) [Info](#)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0fab05a132050006	-	-	Yes	vpc-099792461c7d0a616	46364677...

Last updated 24 minutes ago [Actions](#) [Create route table](#)

Select a route table

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- Enter the name: Public Route Table.
- Associate it with the new VPC.

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional
 [Remove](#)

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create route table](#)

Now open the created Route Table and click on **Edit Routes.**

Route table rtb-035bbbeb0291295af | Public-RT was created successfully.

rtb-035bbbeb0291295af / Public-RT [Actions](#)

Details [Info](#)

Route table ID
[rtb-035bbbeb0291295af](#)

VPC
[vpc-06c203f6d6b025c5b | app-vpc](#)

Main
[No](#)

Owner ID
[\[redacted\]](#)

Explicit subnet associations
-

Edge associations
-

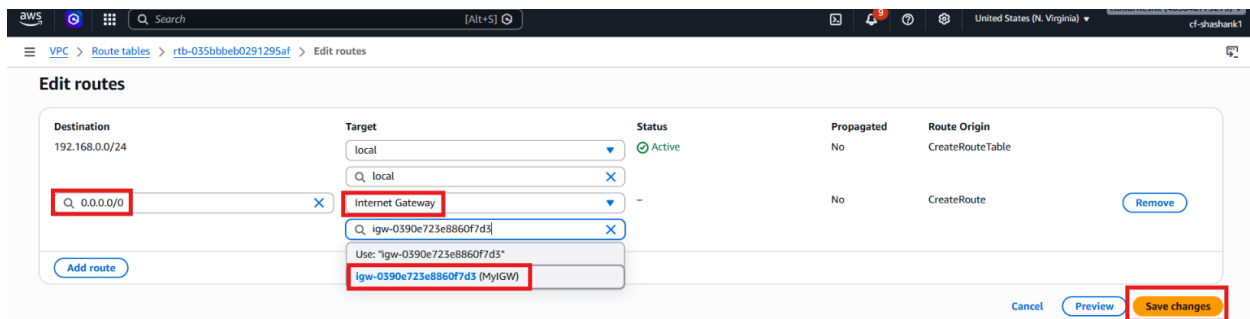
Routes **Subnet associations** **Edge associations** **Route propagation** **Tags** **Related resources**

Routes (1)

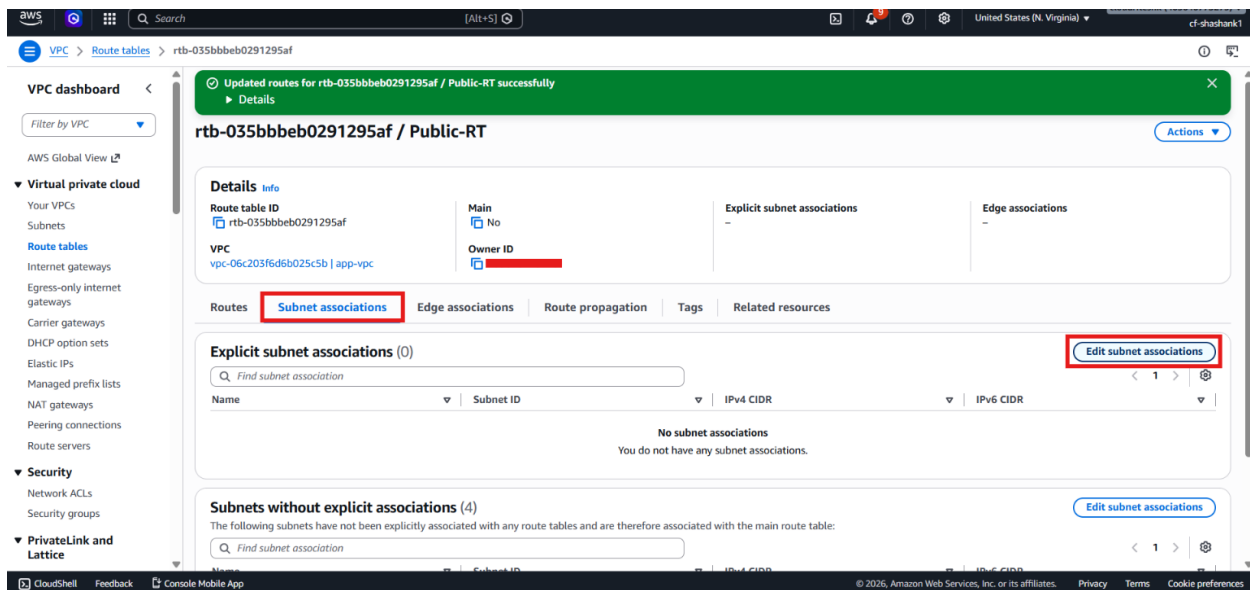
Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	Create Route Table

[Both](#) [Edit routes](#)

- Add the default route 0.0.0.0/0.
- Select the **Internet Gateway** as the target.



- Associate both public subnets with this route table.
- For this click on **Subnet Association** tab and click on **Edit subnet association**.



- Now choose both **Public Subnets** and click on **Save association**.

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (2/4)

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public Subnet 1	subnet-018f92fe3e08eca78	192.168.0.0/26	–	Main (rtb-0bf90b0023a069f19)
☐	Private Subnet 1	subnet-00f3f6aad5a8a847	192.168.0.128/26	–	Main (rtb-0bf90b0023a069f19)
☐	Private Subnet 2	subnet-021cd1fda47875be2	192.168.0.192/26	–	Main (rtb-0bf90b0023a069f19)
☒	Public Subnet 2	subnet-0ac948afb760b6a77	192.168.0.64/26	–	Main (rtb-0bf90b0023a069f19)

Selected subnets

subnet-018f92fe3e08eca78 / Public Subnet 1 subnet-0ac948afb760b6a77 / Public Subnet 2

Cancel **Save associations**

Step 6 – Create the Private Route Table

- Click **Create Route Table**.
- Enter the name Private RT.
- Associate it with the same VPC.

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Private-RT

VPC
The VPC to use for this route table.

vpc-06c203f6d6b025c5b (app-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q. Name Q. Private-RT Remove

Add new tag

You can add 49 more tags.

Cancel **Create route table**

- Add the default route 0.0.0.0/0.
- Select the **NAT Gateway** as the target.

AWS CloudShell Search [Alt+S]

VPC > Route tables > rtb-018456bcb37c86e75 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Use: "nat-"
nat-025b56d78e06fb183 (MyNGW)

Cancel Preview **Save changes**

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- Associate both private subnets with this route table as we did for Public-RT.

AWS CloudShell Search [Alt+S]

VPC > Route tables > rtb-018456bcb37c86e75 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public Subnet 1	subnet-018f92fe5e08eca78	192.168.0.0/26	-	rtb-035bbbeb0291295af / Public-RT
☒ Private Subnet 1	subnet-00ff3f6aad3a8a847	192.168.0.128/26	-	Main (rtb-0bf9060023a069f19)
☒ Private Subnet 2	subnet-021cd1fda47875be2	192.168.0.192/26	-	Main (rtb-0bf9060023a069f19)
Public Subnet 2	subnet-0ac948afb760b6a77	192.168.0.64/26	-	rtb-035bbbeb0291295af / Public-RT

Selected subnets

subnet-00ff3f6aad3a8a847 / Private Subnet 1 subnet-021cd1fda47875be2 / Private Subnet 2

Cancel **Save associations**

Step 7 – Verify the Network Architecture

- Confirm that the VPC contains two public subnets and two private subnets.
- Confirm that the **Internet Gateway** is attached to the VPC.
- Confirm that the **NAT Gateway** is running inside a public subnet.
- Confirm that the public and private route tables are associated with the correct subnets.
- Verify that the network is ready for ECS deployment.

Verification

- Verified that the custom **Amazon VPC** was created successfully.

- Verified that two public subnets and two private subnets were created across two Availability Zones.
- Verified that the **Internet Gateway** was attached to the VPC.
- Verified that the **NAT Gateway** was created with an associated **Elastic IP**.
- Verified that the public route table routed internet traffic through the Internet Gateway.
- Verified that the private route table routed outbound traffic through the NAT Gateway.
- Verified that the correct subnet associations were configured.
- Verified that the secure network environment was ready for configuring security groups and deploying the ECS application.